Troy University - Program at Hanoi University of Science and Technology

[Code 2]

FINAL EXAM QUESTIONS - Fall Semester 2025
Course: Applied Statistics (MTH 210). **Time:** 120 minutes

Instruction: Give answers correct to 3 decimal places.

Question 1 (10 points) A data packet is sent through a network that can operate in two modes: The network is in high-load mode with probability 0.25, whereas the network is in normal-load mode with probability 0.75. Packet loss occurs with probability 0.2 when the network is in high-load mode, or with probability 0.1 when the network is in normal-load mode. Let H be the event that the network is in high-load mode, and L be the event that a packet is lost.
a) Compute the probability that a packet is lost, $\mathbb{P}(L)$.
b) Compute the probability that the network is in normal-load mode given that the packet is not lost, $\mathbb{P}(H^c|L^c)$.

Question 2 (10 points) Khang has three coins. One coin is fair, one coin is biased so that the probability of obtaining a head is $\frac{1}{5}$ and the third coin is biased so that the probability of obtaining a head is $\frac{1}{4}$. The random variable X is the number of heads that Khang obtains when he throws all three coins at the same time. Find the probability mass function (PMF) and the variance of X.

Question 3 (10 points) The length of time in minutes that a customer queues in a bank is a continuous random variable T with probability density function (PDF)
$$f_T(t) = \begin{cases} \frac{1}{144}(36 - t^2) & \text{if } t \in [0, 6], \\ 0 & \text{otherwise.} \end{cases}$$
Find the conditional probability $\mathbb{P}(T \leq 4 | T \geq 1)$.

Question 4 (10 points) Let X and Y be two jointly continuous random variables with joint probability density function (PDF)
$$f_{XY}(x, y) = \begin{cases} \frac{5}{12}x + \frac{1}{4}y^2 & \text{if } 0 \leq x \leq 2,\ 0 \leq y \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$
Find the marginal PDF $f_X(x)$ of X, and the expectation $\mathbb{E}[XY]$.

Question 5 (30 points) A random sample $X_1, X_2, X_3, \ldots, X_{22}$ is given from a normal distribution with unknown mean $\mu = \mathbb{E}[X_i]$ and unknown variance $\mathrm{Var}(X_i) = \sigma^2$. For the observed sample, the sample mean is $\bar{X} = 30.66$, and the sample variance is $S^2 = 3.604$.

a) Find a 99% confidence interval for μ.
b) Find a 95% confidence interval for σ^2.

Question 6 (15 points) A random sample of 14 workers from a mobile phone assembly line is selected from a large number of workers. A manager asks each of these workers to assemble a phone at their normal working speed. The time taken, in minutes, to complete these tasks are recorded below.
$$43.2, 41.6, 49.3, 48.2, 44.2, 40.6, 39.7, 43.4, 44.9, 45.1, 46.2, 43.2, 44.5, 45.5$$
Assuming that this sample comes from a normal distributed population, test the claim at the 5% significance level that the population mean is different from 45 minutes.

Question 7 (15 points) Let $X_1, X_2, X_3, \ldots, X_{120}$ be a random sample from an unknown distribution. After observing this sample, the sample mean is $\bar{X} = 24.2$, and the sample variance is $S^2 = 2.917$. Design a significance level 0.05 test to choose between $H_0 : \mu = 25$, $H_1 : \mu < 25$.

The critical values of standard normal distribution are $z_{0.05} = 1.645$, $z_{0.025} = 1.96$. The critical values of Student's t-distribution are $t_{0.005,21} = 2.831$, $t_{0.01,21} = 2.518$, $t_{0.025,13} = 2.160$, $t_{0.05,13} = 1.771$. The critical values of Chi-Square distribution are $\chi^2_{0.025,21} = 35.479$, $\chi^2_{0.975,21} = 10.283$.